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A lightweight video anomaly detection model with weak supervision and adaptive instance selection

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Highlights

- A new lightweight video anomaly detection model is proposed.
- The Weakly labeled data problem is mitigated by an adaptive sampling strategy.
- A lightweight multi-level temporal correlation attention module is developed.
- A lightweight hourglass-shaped fully connected layer is designed.
- Extensive experiments are conducted, which show that the proposed method is both lightweight and effective.

Abstract

Video anomaly detection is to determine whether there are any abnormal events, behaviors or objects in a given video, which enables effective and intelligent public safety management. As video anomaly labeling is both time-consuming and expensive, most existing works employ unsupervised or weakly supervised learning methods. This paper focuses on weakly supervised video anomaly detection, in which the training videos are labeled whether or not they contain any anomalies, but lack information about the specific frames and quantities of anomalies. However, the uncertainty of weakly labeled data and the large model size prevent existing methods from wide deployment in real scenarios, especially the resource-limit situations such as edge-computing. In this paper, we develop a lightweight video anomaly detection model. On the one hand, we propose an adaptive instance selection strategy, which is based on the model's current status to select confident instances, thereby mitigating the uncertainty of weakly labeled data and subsequently promoting the model's performance. On the other hand, we design a lightweight multi-level temporal correlation attention module and an hourglass-shaped fully connected layer to construct the model, which can reduce the model parameters to only 0.56% of the existing methods (e.g. RTFM). Extensive experiments on three public datasets UCF-Crime, ShanghaiTech and XD-Violence show that our model performs better than or equally to the existing lightweight methods, while with a significantly reduced number of model parameters. Furthermore, by integrating the improved module designed in this paper with the VadCLIP method proposed by Wu et al., we achieve the state-of-the-art performance of non-lightweight models on the UCF-Crime and XD-Violence datasets.

Previous Next

Keywords

Video anomaly detection; Weak supervision; Adaptive instance selection; Lightweight model

1. Introduction

Surveillance serves as a critical tool for identifying unexpected or abnormal events in many scenarios such as traffic monitoring and public safety management. Traditionally, video surveillance is heavily dependent on manual operations and of low intelligence [1], [2], [3]. For example, many cameras are installed in public venues such as stations and parks, to monitor unexpected or abnormal events, which generate huge amounts of videos. To check these videos manually is time-consuming and laborious. The rapid development of computer vision and deep learning technologies has spurred more and more research on video abnormal event detection or video anomaly detection (VAD), which enables the applications of automatic scene monitoring and intelligent early warning.

Generally, there are there types of video anomaly detection methods, supervised [4], unsupervised [5], [6], [7], [8], [9] and weakly-supervised [3], [10], [11], [12], [13], [14], [15]. Supervised video anomaly detection typically requires frame-level or even pixel-level labels, which incurs expensive training cost. Hence, there is a little related research in this direction. Unsupervised video anomaly detection uses unlabeled data to train models, with the lowest training cost, but exhibits poor performance. Weakly-supervised video anomaly detection (WVAD) uses weakly-labeled data to train the model, where the training videos are labeled whether or not they contain any anomalies, but there is no information about which frames the anomalies are located. Thus, WVAD inherits the advantages of supervised and unsupervised methods. On the one hand, WVAD has better performance than unsupervised methods as some supervision is exploited. On the other hand, WVAD is much cheaper and more efficient in acquiring the training data than supervised methods, because the latter requires to label each frame whether it contains anomalies. Therefore, WVAD becomes a hot topic of video anomaly detection.

WVAD is commonly implemented under the framework of Multiple Instance Learning (MIL) [3], [10]. With the MIL paradigm, videos are conceptualized as bags containing multiple clips, each of which is treated as an instance. Annotations for training videos are provided at the video level, indicating which videos contain anomalies without specifying the exact clips (or instances) and frames where anomalies are present. Furthermore, existing models are often considerably large, posing deployment challenges in resource-constrained environments such as edge computing applications. This paper addresses two widely recognized WVAD challenges. The first one concerns the uncertainty of weakly labeled data, i.e., the lack of specific information about the number and locations of anomalous clips within each anomalous video, which limits the effective utilization of training data, consequently leading to unsatisfactory performance [16], [17]. The second one relates to the large scale of models, i.e., the excessive number of parameters in existing methods [14], [18], as mentioned above.

Concretely, existing MIL based methods [3], [10], [11], [12], [18] adopt a strategy to maintain balance between the number of anomalous videos and that of normal videos in the training set, and compute the loss by selecting the instance (or clip) of the highest anomaly score. This strategy aims to minimize the uncertainty of weakly labeled data. Typically, in real scenarios, the time duration of anomalous videos used for training usually exceeds 30 s, which suggests the presence of multiple anomalous clips. However, the selection of only the instance (clip) of the highest anomaly score might lead to under-utilization of all the available anomalous data. Statistical finding from RTFM [13] reveals that the mean number of anomalous clips within an anomalous video is approximately three. Therefore, choosing the three instances (clips) of the largest feature magnitude in both anomalous and normal videos for loss calculation can more effectively utilize the training data, thereby improving performance. However, our point of view is that different numbers of instances should be chosen for different datasets. Therefore, we propose an Adaptive Instance Selection (AIS) strategy, which selects the number of normal or anomalous instances adaptively based on the current status of the model in training, thereby it is able to accommodate different training datasets.

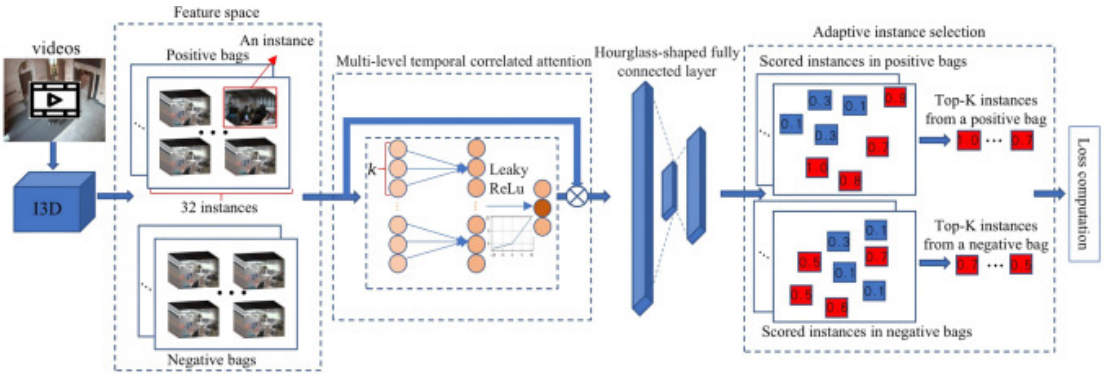

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Fig. 1. The framework of our method Light-WVAD. Our model is based on the multi-instance learning (MIL) framework. Each video is divided into 32 consecutive clips (or instances), which are grouped into a positive instance bag (for abnormal videos) or a negative instance bag (for normal videos). Video features are extracted by I3D. A Multi-level Temporal correlation Attention (MTA) module is designed to capture time-related information, which is then input to a Hourglass-shaped Fully Connected layer (HFC) to calculate the score of each instance. The top-K reliable instances are selected based on an Adaptive Instance Selection (AIS) strategy for subsequent loss calculation.

On the other hand, there are relatively fewer lightweight models for VAD in the literature [14], [15], [18], which usually compromise performance to embrace smaller models. In this paper, to devise a lightweight yet high-performance anomaly detection model, we develop a lightweight Multi-level Temporal correlation Attention (MTA) module. This module emphasizes the relationships between video instances (clips) of different time

spans, thus making the model focus on the important instances in the videos. Furthermore, we design an Hourglass-shaped Fully Connected layer (HFC), which contains only half the parameters of a conventional fully connected layer (FC), yet does not degrade the model's performance. Finally, Observing the limitation of the sparsity loss widely used in existing methods, we specifically introduce a more effective antagonistic loss.

In summary, in this paper we propose a novel, MIL-based lightweight video anomaly detection model, whose framework is shown in Fig. 1. First, we use I3D to decompose each video into 32 continuous clips, which are encoded and organized into the positive or negative instance bag. The instances in the bags are input into the Multi-level Temporal correlation Attention (MTA) module, to make the model focus on the important features in the time dimension of the videos. Next, instance features are input into an Hourglass-shaped Fully Connected (HFC) layer to obtain the anomaly score for each instance. Finally, we dynamically determine the number K of positive and negative instances used for loss calculation based on the adaptive instance selection strategy. The main contributions of this study include:

1. We propose an adaptive sample selection strategy that alleviates the uncertainty problem of weakly-labeled data and improves model performance.
2. We design a lightweight multi-level temporal correlation attention module and an hourglass-shaped fully connected layer, which leads to a lightweight video anomaly detection method with only 0.56% of the parameters of the existing well-known RTFM method [13].
3. We analyze the limitations of using sparsity loss in weakly-supervised video anomaly detection, and develop a more suitable antagonistic loss for this problem.
4. We conduct extensive experiments on three benchmark datasets UCF-Crime, ShanghaiTech and XD-Violence, which show that our model achieves superior or equal performance but much less parameters compared to the existing lightweight models. Further combining our improved module with the existing VadCLIP method, we achieve the state-of-the-art performance of non-lightweight models on the UCF-Crime and XD-Violence datasets.

The rest of this paper is organized as follows: Section 2 reviews the related works and highlights the differences of our work from the existing ones. Section 3 introduces the proposed method in detail. Section 4 presents the results of performance evaluation, including performance comparison with existing works and ablation studies. Finally, Section 5 concludes the paper and pinpoints the future works.

2. Related work

Here we review the related work from two aspects: weakly supervised video anomaly detection and attention network.

2.1. Weakly supervised video anomaly detection

Traditional anomaly detection methods typically assume that only normal training data is available and use hand-crafted features for one-class classification to solve the problem [19], [20]. With the development of deep learning techniques, some unsupervised learning methods utilize deep neural networks to learn features such as human posture and optical flow [5], [21], [22], [23], or utilize the difference in feature distribution between normal and abnormal samples [9] to perform abnormality detection. The essence of unsupervised video anomaly detection methods [5], [21], [22], [23] lies in the assumption that anomalies are rare events, and therefore the model is trained to learn mainly the feature distribution of normal samples. Based on this assumption, the model then determines whether a test sample is an anomaly by calculating the reconstruction errors or feature distributions of the test sample and the normal samples. However, due to the lack of prior

knowledge about anomalies, these methods are prone to overfitting to training data and unable to distinguish normal and anomalous events.

Some works [3], [10], [11], [12], [13] have shown that utilizing partially labeled abnormal samples can achieve better performance than unsupervised methods. However, the cost of obtaining a large number of frame-level labels is prohibitively high. Therefore, some video anomaly detection methods apply video-level labels for weakly supervised training. Sultani et al. [3] proposed a method that uses video-level labels and introduced the large-scale WVAD dataset UCF-Crime. This makes WVAD one of the mainstream research directions in video anomaly detection [4], [16], [24], [25], [26], [27], [28].

Existing Weakly supervised video anomaly detection methods are mainly based on multiple-instance learning (MIL). Given that the training data possesses only video-level labels, these approaches typically employ the instances of the highest anomaly prediction score from both positive and negative bags for loss computation during training, leading to under-utilization of the training data. To address this issue, Zhong et al. [29] transformed WVAD into a binary classification problem in the presence of label noise. They employed a Graph Convolutional Neural Network (GCN) [30] to eliminate label noise, thus enhancing data utilization and model performance. While this method improves model performance, the training computation cost associated with GCN and MIL is considerably high. Furthermore, it may make features unconstrained in the feature space, resulting in unstable performance. Furthermore, Tian et al. [13] amalgamated representation learning and anomaly score learning by devising Robust Temporal Feature Magnitude (RTFM) learning. They separately selected the three instances of the largest feature magnitude, in both abnormal and normal bags for loss calculation. Such an approach can effectively utilize the training data, thus achieving better performance. However, this training data utilization strategy does not consider the diverse characteristics of different datasets.

To address the issue of large model parameters, Chang et al. [14] proposed a lightweight multi-instance learning (MIL) model that incorporates a comparative attention module to enhance performance. Additionally, Watanabe et al. [15] adopted a self-attention mechanism, which further improves the effectiveness of the lightweight MIL model. BE-WVAD [18] introduces a short input inference mode, significantly reducing the required input video length, thereby lowering memory and computational costs. The knowledge distillation technique proposed by Hinton et al. [31] aims to transfer knowledge from a large teacher model to a smaller student model by aligning their probabilities, thereby enhancing the performance of the latter. Liu et al. [26] designed an effective multimodal learning strategy and a distillation loss function based on distillation learning, allowing for the use of a single modality during the inference phase while maintaining high performance in anomaly event detection. The works above have effectively addressed the challenges posed by large model parameters while lifting the performance and efficiency of models in practical applications. However, there is still much space to improve performance and cut off model size for real applications.

2.2. Attention network

Attention networks are initially used for machine translation [32] and later widely used for various computer vision tasks such as image classification [33], object detection [34], image segmentation [35], image captioning [36], and action recognition [37] etc., and have achieved excellent performance. Recently, attention networks have also been employed in weakly supervised tasks. Choe et al. [38] proposed an attention-oriented dropout layer, leveraging self-attention mechanisms to address the weakly supervised object localization problem. W-TALC [39] combines MIL and common activity similarity loss to train an attention module to solve the weakly supervised action localization problem in videos. Zhou et al. [7] proposed an attention that focuses on the foreground of an image to alleviate the foreground-background imbalance problem in anomaly detection. Li et al. [40] improved the ability of weakly supervised anomaly detection models to extract relationships between video frames by using SE-attention [41]. However, this method, not specifically designed for video data, fails to

enable the model to concentrate on the temporal correlation between consecutive instances (clips), and has a relatively large number of parameters.

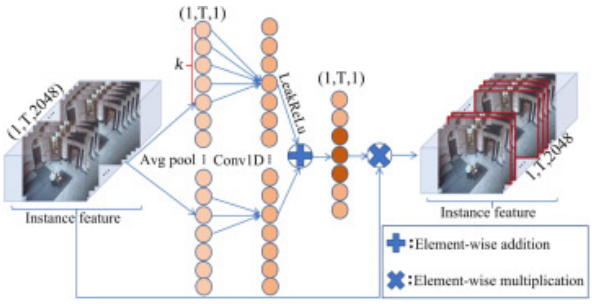

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Fig. 2. The structure of the multi-level temporal correlation attention (MTA) module. Here, each video is divided into T (32 in this paper) clips, each of which corresponds to an instance.

3. Methodology

This section presents the proposed Lightweight Weakly Supervised Video Anomaly Detection (Light-WVAD in short) method. The framework of Light-WVAD is shown in Fig. 1. Light-WVAD mainly comprises a lightweight multi-level temporal correlation attention module, an hourglass-shaped fully connected layer, and an adaptive instance selection strategy to alleviate the uncertainty of weakly labeled data. In addition, we employ a more robust antagonistic loss to further optimize the model's performance. In what follows, we first give a formal definition of the problem, then introduce the major modules of our method in detail.

3.1. Problem statement

In the context of weakly-supervised learning, anomaly detection can be typically regarded as a multiple-instance learning problem. Given a set of videos with video-level annotations (i.e., just labeling whether or not a video in the training dataset contains anomalous content), WVAD aims to train a model with the annotated data, which is able to predict whether there are anomalies in any new videos.

3.2. Feature extraction

In the data preprocessing stage, the widely used networks for feature extraction are I3D [42] and C3D [43]. Some existing research works [13], [16], [17] have shown that I3D can more effectively extract sample features. Therefore, in this paper we use I3D for data preprocessing and convert videos into feature vectors. Concretely, Given a video V_i, we divide it into 32 consecutive and non-overlapping clips, each of which is regarded as an instance. The clips are grouped into a positive or negative bag based on the video-level labels Y. Here, the positive bag ($Y = 1$) contains at least one anomalous instance (clip), while the negative bag ($Y = 0$) consists only of normal instances (clips).

3.3. Multi-level temporal correlation attention

The structure of the Multi-level Temporal Correlation Attention (MTA) module is illustrated in Fig. 2. In this module, we initially employ a global average pooling layer, which serves to transform the T (T is set to 32 for the experiments presented in this paper) instance features within a bag into a T-dimensional vector, thereby encapsulating the essence of each channel's temporal characteristics. This transformation is critical as it condenses the spatial and temporal information into a more manageable and representative form. Subsequently, to exploit the temporal correlations within the video data, we utilize convolution operations to facilitate feature interactions between neighboring channels. This step is essential for deriving a temporal

attention mechanism that assigns a weight to each channel based on its relevance within the temporal sequence. By doing so, the model is enabled to selectively emphasize the significance of individual instances, thereby making it focusing on the most informative segments of each video in the temporal domain.

In order to maintain model efficiency and reduce the number of parameters, we adopt a one-dimensional convolutional layer ($Conv1D$) with a kernel size of k ($k \geq 3$). This choice of $Conv1D$ is important as it effectively captures the cross-channel interactions of adjacent instances while keeping the computational overhead acceptable. Furthermore, to comprehensively capture the interaction information across varying temporal scales, we employ convolution kernels of diverse sizes. This multi-scale approach allows the model to attend to both fine-grained and coarse-grained temporal dependencies, providing a more robust representation of the temporal dynamics. The computation process is formulated as follows:

$$T_{attention} = (Conv1D\,[k]\,, Conv1D\,[k-2]\,, \ldots, Conv1D\,[3]) * G\,(\chi) \tag{1}$$

where $Conv1D\,[k]$ denotes the convolution kernel function of size k ($k \leq T$), $*$ is the convolution operation. The variable χ stands for the feature vector of a video, while $G\,(\chi)$ represents the feature vector derived from the global pooling process. This T-dimensional vector is processed by multiple convolutional layers, each of which is equipped with different kernel sizes to extract unique temporal patterns. The convolution operation within the one-dimensional convolution layer, involving $Conv1D\,[k]$ and $G\,(\chi)$, is analogous to sliding the kernel across the feature vector, executing element-wise multiplication, and summing the results to produce a weighted feature representation that emphasizes the relevant temporal interactions. The concatenated feature maps from these convolutions form a comprehensive, multi-dimensional representation that captures a range of temporal correlations. The calculation of $G\,(\chi)$ is as follows:

$$G\,(\chi) = \frac{1}{WH} \sum_{i=1,j=1}^{W,H} \chi_{ij} \tag{2}$$

where W and H denote the width and height of the input feature matrix, respectively. In the final stage of the attention module, the computed temporal attention weights $T_{attention}$ are processed by an activation layer, specifically the Leak Relu proposed by Glorot et al. [44]. This activation function introduces a small positive slope in the negative part of the function, which helps to mitigate the vanishing gradient problem and allows for better training of deep neural networks. Subsequently, the output of the Leak Relu activation is subject to a summation operation and then concatenated with the original input feature vector $G\,(\chi)$. This concatenation operation serves to integrate the attention-weighted features with the initial feature representation. The output $Y_{attention}$ of the attention module is

$$Y_{attention} = \lambda_1 Sum\,(L_Relu\,(T_{attention})) \cdot \chi \tag{3}$$

Here, $L_Relu\,()$ [44] represents the activation function, and λ_1 is a hyperparameter that is empirically set to 0.1 to control the scaling of the attention mechanism's impact on the feature representation. $\cdot$ denotes the element-wise multiplication operation that combines the activated and summed attention weights with the original feature vector χ. The resulting $Y_{attention}$ constitutes the video feature with integrated multi-level temporal correlations obtained by MTA. This refined feature representation amplifies the features of important instances within each video while restraining the features of less significant instances, thereby enhancing the overall discriminative power of the model.

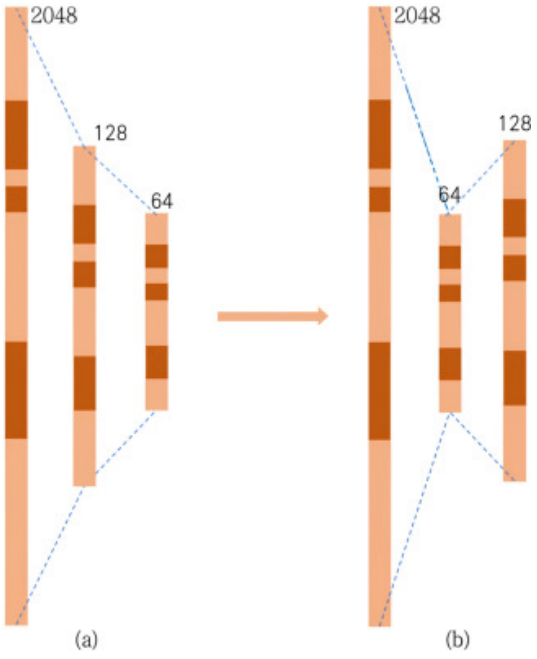

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Fig. 3. The structures of (a) the traditional fully connected layer (FC), and (b) our hourglass-shaped fully connected layer (HFC).

3.4. Hourglass-shaped fully connected layer

In the fully connected layer, each node connects each node of the preceding layer, which leads to a high number of parameters in this layer. To address the parameter redundancy, we introduce a novel hourglass-shaped fully connected (HFC) layer. As shown in Fig. 3, the structure of HFC on the right side is compared with the traditional fully connected (FC) layer on the left side. Despite HFC and FC have the same number of layers and layer dimensions, HFC adopts a *2048-64-128* configuration, in contrast to the conventional FC's *2048-128-64* arrangement. This structural difference results in that HFC has approximately half the number of parameters of the FC layer. With the HFC layer, we can derive the anomaly scores for all instances within both the positive and negative bags.

HFC is composed of three layers, where the initial layer receives input from 2048 neurons and transmits to 64 neurons. To analyze the computational complexity of HFC, we consider the dimensions of the weight matrix. For the first layer of HFC architecture, the weight matrix is 2048 × 64, which necessitates 131,072 multiplication operations. Furthermore, it requires 64 additional operations, resulting in a total of 131,132 operations for the first layer. The subsequent layer processes input from 64 neurons and outputs to 128 neurons, requiring 8192 multiplication operations and 128 addition operations, totaling 8,320 operations for this layer. The combined computation count is 139,452. As a comparison, the total computation cost for a traditional fully connected layer (2048 × 128 × 64) is 272,512 operations. This indicates that HFC can significantly reduces computational overhead while maintaining the functionality of the network.

3.5. Adaptive instance selection

Here, we design an adaptive instance selection (AIS) strategy to choose important instances based on the current training status of the model and autonomously determines the number of instances used for loss computation. This strategy is primarily based on two important facts of MIL-based weakly supervised video anomaly detection. First, the negative bag contains only normal instances, i.e., each instance in the negative bag is normal. Second, as continuity exists amongst instances in the videos, the predicted anomaly scores of neighboring instances should also be continuous. These two factors enable us to infer the model's training

status by examining its predictions of negative instances during the training process. With this, the confident positive (anomalous) instances can be determined from the set of current positive instances.

The workflow of AIS is shown in Fig. 4, which consists of three steps:

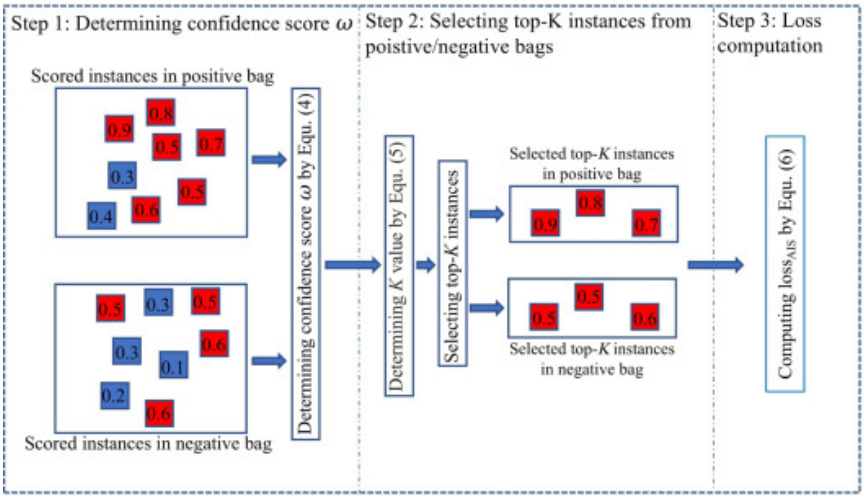

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Fig. 4. The workflow of our adaptive instance selection (AIS) strategy on a pair of positive and negative bags. It consists of three steps. Here, each red or blue square is an instance. Top-K instances are selected from both the positive and negative bags.

Step 1. With the anomaly scores of all instances from the HFC module, AIS calculates the confidence score ω that measures the mature degree of the model (i.e., how well the model is trained), based on the average anomaly score of the negative instances and the average score difference between consecutive positive and negative instances. Concretely,

$$\omega = 1 - \frac{1}{T}\sum_{i=1}^{T} S_i^N - \frac{1}{2T-2}\sum_{i=1}^{T-1}\left(\left|S_{i+1}^N - S_i^N\right| + \left|S_{i+1}^P - S_i^P\right|\right) \tag{4}$$

Above, T is the number of instances in each positive/negative bag, S^P and S^N are the scores of positive and negative instances respectively. In Eq. (4), the 2nd item computes the mean of anomaly scores of negative instances, while the 3rd item calculates the mean of the differences in anomaly scores between consecutive positive and negative instances. As training goes, the 2nd term approaches 0 because negative instances should have 0 anomaly score, and the 3rd term is also close to 0 considering the continuity of anomaly scores of consecutive instances. Thus, the confidence score ω becomes close to 1.

Step 2. We select confident instances for loss computation based on the confidence score ω. The number of confident instances to be selected is determined as follows:

$$K = \omega * \sum_{i=1}^{T} f\left(S_i^P\right) \tag{5}$$

where the function $f(.)$ is 1 if $S_i^P \geq 0.9$, otherwise is 0. That is, we first count the number of instances with score over 0.9, and then multiply it by the confidence score ω to obtain the final number of confident anomaly instances K. Considering the possibility that the ω value is less than 0 during training, the minimum value of K in this paper is set to 1.

Step 3. The feature magnitudes [13] of the top-K instances with the highest scores are selected separately from the positive and negative sets. Thus, the loss function is as follows:

$$loss_{AIS} = \sum_{\chi \in U_{\chi_top-K}} \Big(y log\left(mean\left(s_\theta\left(\chi\right)\right)\right) + \left(1-y\right) log\left(1 - mean\left(s_\theta\left(\chi\right)\right)\right)\Big) \tag{6}$$

where s_θ is the feature extraction part of the model, $mean$ is the mean function. U is the set of video instance (clip) features extracted by I3D, and U_{χ_top-K} is the set of features of the top-K instances selected from the positive and negative bags, χ is the feature of an instance. $y \in \{0, 1\}$, and $y = 0$ when χ is a normal video clip feature, otherwise $y = 1$.

3.6. Antagonistic loss function

Existing methods for weakly supervised anomaly detection predominantly rely on smooth loss and sparsity loss for model optimization. The smooth loss is based on the assumption that there is a minimal discrepancy in features between consecutive instances, leading to a negligible variation in anomaly scores, which conforms to real-world scenarios. Conversely, the sparsity loss is based on the premise that anomaly events in videos are rare, suggesting that the mean anomaly score of instances within an anomalous video should be close to zero. However, this sparsity assumption, while reasonable in theory, does not hold true during actual model training due to computational constraints. Typically, a bag consists of 32 consecutive instances. In cases where anomalies are present within a bag, the proportion of abnormal instances cannot be disregarded. Our preliminary analysis on the ShanghaiTech dataset [29] reveals that the ratio of abnormal instances is approximately 20%, with a mean score of around 0.2. Based on this analysis, we use only the smoothing loss in our method. Concretely, the smoothing loss is defined as follows:

$$loss_{smooth} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\|S_{i+1} - S_i\|^2 \right) \tag{7}$$

where S represents the anomaly scores predicted by the model for instances, and $\|\|^2$ is the $L2$ norm, which assigns larger losses and gradients to anomalous instances.

In addition to the smoothing loss, we develop an antagonistic loss that is particularly suitable for weakly supervised Multiple Instance Learning (MIL). The antagonistic loss is grounded on the assumption of quite different scores for positive and negative instances, which allows for an assessment of the model's predictive performance on both types of instances. Given that the negative bag contains only normal instances, the anomaly scores of the least anomalous instances should tend towards zero. Conversely, for the positive bag, which includes some abnormal instances, the anomaly scores of the most anomalous instances should be close to 1. Based on this observation, we formulate the antagonistic loss as follows:

$$loss_{antagonistic} = S^N_{top-1} + \left(1 - S^P_{top-1}\right) \tag{8}$$

Here, S^P_{top-1} and S^N_{top-1} respectively represent the highest (top-1) abnormality score of instances in the positive and negative bags predicted by the model. $loss_{antagonistic}$ not only implies the antagonistic relationship between positive and negative instances but also provides specific optimization constraints for them. Ultimately, the total loss function of our proposed method, Light-WVAD, is a combination of the anomaly instance selection (AIS) loss, the smoothing loss, and the antagonistic loss as follows:

$$loss_{all} = loss_{AIS} + loss_{smooth} + loss_{antagonistic} \tag{9}$$

4. Performance evaluation

4.1. Datasets and evaluation metrics

We evaluate our model on two commonly used video anomaly detection benchmark datasets, ShanghaiTech [45] and UCF-Crime [3]. The videos in both datasets were collected by fixed recording devices (surveillance

cameras).

1. ShanghaiTech [45]: This dataset consists of 437 street surveillance videos captured from fixed angles, with 13 different background scenes. It contains 307 normal videos and 130 abnormal videos. Originally, it was used as a benchmark for unsupervised video anomaly detection. However, Zhong et al. [29] reorganized the dataset by selecting a subset of anomalous testing videos to create a weakly supervised training set, so that all 13 background scenes are covered in both the training and testing sets. We follow exactly the same procedure as in Zhong et al. [29] to convert ShanghaiTech for the weakly supervised setting.

2. UCF-Crime [3]: The UCF-Crime dataset is a real-world surveillance video dataset that consists of 950 anomalous videos belonging to 13 anomalous categories, and 950 normal videos. The training set provides only video-level labels, while the testing set provides both video-level labels and frame-wise annotations for evaluation. Each anomalous video in the testing set contains one or two anomalous events. Due to significant variations in the time duration of abnormal events across different videos, this dataset poses a serious challenge for video anomaly detection models.

3. XD-Violence [46]: XD-Violence is the largest publicly available multimodal dataset for violence detection. It consists of 4754 untrimmed videos with a total duration of 217 h, covering six types of violence and providing RGB, optical flow, and audio streams for each video. The dataset is collected from movies and YouTube, encompassing diverse scenarios. The entire dataset is divided into a training set of 3954 videos with video-level labels and a test set of 800 videos with frame-level labels.

This study's performance evaluation uses frame-based receiver operating characteristic (ROC) curves and the corresponding area under the curve (AUC), as well as established methodologies in previous studies [3], [14], [15], [47], [48]. Specifically, the UCF and ShanghaiTech datasets rely on AUC for evaluation. Additionally, XD-Violence employs frame-level average precision (AP) as its evaluation metric.

4.2. Compared methods

We compare our method with a number of existing works, including thirteen general (non-lightweight) models and three lightweight methods. For some models, we consider two versions of using different feature extractors.

Among the thirteen non-lightweight models, the work of Sultani et al. [3] is the first to use the MIL framework and weakly supervision for video anomaly detection, and most of the subsequent works follow their settings. The method of Zhang et al. [10] defines a new inner-bag loss in the MIL framework to limit the function space and expand the differences in feature distributions for different types of instances. ARNet [12] uses a dynamic multi-instance learning loss to enlarge the interclass distance between anomalous and normal instances, and a center loss to narrow the intraclass distance of normal instances to boost the model's ability to distinguish anomalies. CLAWS Net+ [49] employs a clustering loss to mitigate labeling noise, which improves the learning of representations for both abnormal and normal videos. MIST [17] adopts a multi-instance self-training framework to effectively refine task-specific discriminative representations using only video-level annotations. Similarly, MSL [16] proposes a self-training strategy and designs a transformer-based Multi-Sequence Learning network to further learn video-level anomaly probabilities and clip-level anomaly scores. RTFM [13] trains the feature magnitude learning function to effectively identify positive instances and improves the robustness of the model. Purwanto et al. [4] employs a new relation-aware feature extractor to learn multi-scale features and then uses a conditional random fields (CRFs) with a newly devised self-attention to model the relationships between global and local features. DAR [47] and the method of Wu et al. [11] design the model structure using a

multi-branching and multi-stage approach, respectively, to improve the model's ability to understand and fuse different modal data in the videos. LSTC [25] employs two tubelet-based spatio-temporal transformer networks to learn from short- and long-term video clips, which are alternatively and iteratively trained via a co-teaching strategy, enhancing the ability to handle subtle and varying-duration anomalies through its transformer architecture and long-short temporal co-teaching approach. Mu et al.'s method [50] and GCN-Anomaly [29] innovatively employ graph convolutional networks for enhancing the model's ability to understand spatio-temporal correlation features in videos. The DPK_WSVAD [26] employs distillation learning to design an effective multimodal learning strategy and a distillation loss function, enabling the use of a single modality during the inference phase while maintaining high performance in anomaly event detection. MGFN [51] proposes a novel glance and focus network to effectively integrate spatial–temporal information for accurate anomaly detection. In addition, MGFN [51] adopts the Feature Amplification Mechanism and a Magnitude Contrastive Loss to enhance the discriminativeness of feature magnitudes for detecting anomalies. NG-MIL [52] encodes diverse normal patterns from noise-free normal videos into prototypes to construct a similarity-based classifier. By combining predictions from classifiers, this approach can refine anomaly scores, reducing training instability from weak labels. CU-Net [53] introduces an enhanced framework with a two-stage self-supervised model. This model can generate and iteratively refine pseudo-labels by leveraging the completeness and uncertainty of weakly labeled data. HSN [54] proposes a Human-Scene Network to learn discriminative representations by capturing both subtle and strong cues in a dissociative manner. Despite their remarkable performance in anomaly detection, these models do not consider factors such as model parameter size and runtime in their designs.

As for the three lightweight methods, the work of Chang et al. [14] proposes a lightweight MIL model incorporating a comparative attention module to improve model performance. The method of [15] introduces a self-attention mechanism to enhance the performance of the lightweight MIL model. BE-WVAD [18] proposes short-input inference modes, which can significantly reduce the required length of input videos and therefore greatly cut down memory and computational costs. Although these two methods achieve considerable advance, there remains substantial room to enhance the design of the model and boost the performance.

Table 1. Performance comparison of frame-level AUC on ShanghaiTech. For lightweight models, the best result is in red and the second best in blue; For general (Non-lightweight) models, the best result is in bold.

Model type	Method	Feature extractor	AUC (%)
	GCN-Anomaly [29]	TSN	84.4
	AR-Net [12]	I3D	91.2
	CLAWS Net+ [49]	C3D	89.7
	MIST [17]	C3D	93.1
	MIST [17]	I3D	94.8
	RTFM [13]	C3D	91.5
	RTFM [13]	I3D	97.2
	MSL [16]	C3D	94.8
	MSL [16]	I3D	97.3
	Purwanto et al. [4]	Relation-aware	96.9
	Mu et al. [50]	I3D	92.3
	DPK_WS-VAD [26]	I3D	95.5
General models	DAR [47]	I3D	97.5

Model type	Method	Feature extractor	AUC (%)
	LSTC [25]	I3D	**97.9**
	NG-MIL [52]	I3D	97.4
	HSN [54]	I3D	96.2
	VPE-WSVAD [55]	I3D	96.9
	RTFM[a]	I3D	97.7
	Watanabe et al. [15]	I3D	95.7
	Chang et al. [14]	C3D	87.3
	Chang et al. [14]	I3D	92.3
	BE-WVAD [18]	I3D	95.0
Lightweight models	Light-WVAD (ours)	I3D	95.9

a Represent improved models that combine our proposed adaptive instance selection strategy with an adversarial loss function.

Table 2. Performance comparison of frame-level AUC on UCF. For lightweight models, the best result is in red and the second best in blue; For general (Non-lightweight) models, the best result is in bold.

Model type	Method	Feature extractor	AUC (%)
	Sultani et al. [3]	TSN	75.4
	Zhang et al. [10]	TSN	78.7
	Wu et al. [11]	I3D	82.4
	GCN-Anomaly [29]	TSN	82.1
	CLAWS Net+ [49]	C3D	83.4
	MIST [17]	C3D	81.4
	MIST [17]	I3D	82.3
	RTFM [13]	C3D	83.3
	RTFM [13]	I3D	84.3
	MSL [16]	C3D	82.9
	MSL [16]	I3D	85.3
	Purwanto et al. [4]	Relation-aware	85.0
	Mu et al. [50]	I3D	84.2
	DPK_WS-VAD [26]	I3D	85.4
General models	DAR [47]	I3D	85.2
	LSTC [25]	I3D	85.9
	CU-Net [53]	I3D	86.2
	MGFN [51]	I3D	87.0

Model type	Method	Feature extractor	AUC (%)
	HSN [54]	I3D	85.5
	VadCLIP [56]	CLIP	88.0
	VadCLIP[a]	CLIP	**88.1**
	RTFM[a]	I3D	84.3
	Watanabe et al. [15]	I3D	84.7
	Chang et al. [14]	C3D	83.4
	Chang et al. [14]	I3D	84.6
	BE-WVAD. [18]	I3D	84.1
Lightweight models	Light-WVAD (ours)	I3D	84.7

a Represent improved models that combine our proposed adaptive instance selection strategy with an adversarial loss function.

Table 3. Performance comparison of frame-level AP on XD-Violence. For lightweight models, the best result is in red and the second best in blue; For general (Non-lightweight) models, the best result is in bold.

Model type	Method	Feature extractor	AP (%)
General models	Sultani et al. [3]	C3D	73.2
	Wu et al. [11]	I3D	75.4
	RTFM [13]	C3D	75.9
	RTFM [13]	I3D	77.8
	MSL [16]	C3D	75.5
	MSL [16]	I3D	78.3
	DAR [47]	I3D	78.9
	MGFN [51]	I3D	79.2
	DPK_WS-VAD [26]	I3D	79.0
	NG-MIL [52]	I3D	78.5
	CU-Net [53]	I3D	78.7
	VadCLIP [56]	CLIP	84.5
	VadCLIP[a]	CLIP	**84.9**
Lightweight models	Chang et al. [14]	I3D+flow	71.5
	Chang et al. [14]	I3D	76.9
	BE-WVAD [18]	I3D	74.9
	Light-WVAD(ours)	I3D	77.3

a

Represent improved models that combine our proposed adaptive instance selection strategy with an adversarial loss function.

4.3. Implementation details

Following the experimental setup in [3], we first divide each video into 32 ($T = 32$) video clips. For the model parameters, we have the following settings:

1. The number of nodes in the fully connected (FC) layer is set to 2048, 64, and 128, respectively. The thresholds of the Leacky ReLU activation function and dropout function are set to 0.5.
2. The input features are extracted from the "mix 5c" layer of the pre-trained I3D [42] network, and the multi-level temporal correlation attention module is used with a 1 × 1 Conv1D.
3. Our model is trained end-to-end using the Adam optimizer [57], with a weight decay coefficient of 0.0005, a batch size of 32, and a total of 200 epochs. The learning rate for both ShanghaiTech and UCF-Crime is set to 0.001. Each batch consists of 32 normal and abnormal instances respectively.

We implement our model using PyTorch. To ensure fairness in the comparison of model performance, we use the same benchmark settings as [3], [14], [15], and report the results of all baselines with the same backbone network as ours.

4.4. Performance comparison

To demonstrate our method's effectiveness, we compare it with existing anomaly detection methods, including general WVAD models and lightweight WVAD methods.

Table 1 presents the frame-level AUC results on the ShanghaiTech dataset. We can see that our method obtains the highest performance of 95.9% among the lightweight weakly-supervised methods, and surpasses most of general (non-lightweight) methods, whilst maintaining a parameter size of merely 0.14M.

Table 2 gives the frame-level AUC performance on the UCF dataset. Our method not only achieves the highest AUC of 84.7% among the lightweight weakly-supervised methods, but also outperforms most non-lightweight weakly supervised methods.

Table 3 gives the frame-level AUC performance on the XD-Violence dataset. Our method not only achieves the highest AUC of 77.3% among the lightweight weakly-supervised methods, but also performs comparably to most of the non-lightweight weakly-supervised methods.

To comprehensively assess the robustness and efficacy of the adaptive instance selection (AIS) and adversarial loss functions introduced in this study, we have conducted a series of comparative experiments, with the results being presented Table 1, Table 2, Table 3. We select RTFM and VadCLIP as our baseline models, and integrate the AIS and antagonistic loss functions proposed in this study into them to augment their performance. The corresponding enhanced models are denoted by RTFM* and VadCLIP*. On the ShanghaiTech dataset, RTFM* achieves a 0.5% increase in AUC. This improvement not only verifies the efficacy of the AIS module but also underscores the potential of the proposed adversarial loss function in enhancing model performance. However, on the UCF dataset, RTFM* performs comparably to the original model. This can be primarily attributed to the fact that the TOP-K strategy of the original RTFM model has been optimized specifically for the UCF dataset, leading to a performance that is already close to the best possible on this dataset. While this dataset-specific optimization may not be directly generalizable to other datasets. The AIS module in this study is specifically designed to overcome such limitation. For VadCLIP*, the experimental results indicate that VadCLIP* obtains a performance improvement of 0.1% AUC and 0.4% AP on the UCF and

XD-Violence datasets, respectively. This shows the contribution of the proposed module to enhancing the model's performance across diverse datasets. In conclusion, these comparative experiments confirm the robustness and efficacy of the proposed AIS and adversarial loss functions in enhancing the performance of both RTFM and VadCLIP models across various datasets.

In addition to comparing the AUC performance of different methods, we also compared the number of parameters and the floating-point operations (FLOPs) of several methods, as shown in Table 4. It can be seen from the table that our method only uses 0.14M parameters, which is the least among all the compared models. Specifically, the number of parameters in our method is less than 1% of that in the RTMF method and only half of the currently least resource-intensive lightweight method [14]. In the comparison of FLOPs, our method also achieves the lowest FLOPs value, further demonstrating the fastest computational speed of our method.

Combined the comparison results from Table 1, Table 2, Table 3, Table 4, it is obvious that our method is a lightweight yet good-performance method for weakly supervised video anomaly detection.

Table 4. Model size comparison among weakly supervised methods. The best result is in red and the second best is in blue. M for million, G for billion.

Method	#Parameters (M)	#FLOPs (M)
Sultani et al. [3]	2.11	106
Wu et al. [11]	0.76	133
RTFM [13]	24.72	791
Mu et al. [50]	13.20	–
BE-WVAD [18]	2.49	–
Watanabe et al. [15]	0.33	136
Chang et al. [14]	0.26	175
VPE-WSVAD [55]	>100	>18(G)
Light-WVAD (ours)	0.14	80

4.5. Ablation studies

Effect of major modules in Light-WVAD. To evaluate the effectiveness of the major modules in our method, we conduct ablation experiment on the ShanghaiTech dataset. The results are presented in Table 5. Here, the baseline model is a simple network consisting of fully connected layers, with an AUC of 93.8%. The results in Table 5 show that employing MTA, AIS, and the antagonistic loss (A-Loss in short) to the baseline individually can obviously improve the model's performance. Furthermore, our proposed lightweight HFC structure does not have negative impact on the model's performance. In summary, by combining MTA, HFC, AIS, and A-Loss into our method, a 2.1% performance improvement on the ShanghaiTech dataset is achieved.

Effect of parameter k in MTA. MTA is to capture multi-level temporal correlations of consecutive instances by integrating inter-instance feature relationships across multiple time intervals. As described in Section 3.3, MTA has a hyperparameter k, which represents the maximum number of consecutive k instances, from which MTA can extract temporal correlation information. The value of k affects the model's performance. We change the value of k from 3 to 15 with a stepsize of 2, and report the performance results in Table 6. We can see that when k is set to 5, the model achieves the best performance. By analyzing the performance change when setting different values of k for MTA, we can see that abnormal events typically have a relatively short duration. When k is too large, the proportion of normal instances is too high, which may dampen the influence of

abnormal instances so that MTA fails to provide discriminative temporal correlation information. On the other hand, a too small k may result in insufficient coverage of anomalous instances, so that MTA cannot obtain effective temporal correlation information by enough inter-instance information.

Table 5. Ablation study on ShanghaiTech.

Baseline	MTA	HFC	AIS	A-Loss	AUC (%)-SH
✓					93.8
✓	✓				94.8
✓		✓			93.9
✓	✓	✓			94.9
✓			✓		95.0
✓		✓	✓		95.1
✓	✓		✓		95.4
✓	✓	✓	✓		95.4
✓				✓	95.4
✓	✓	✓	✓	✓	95.9

Effect of the antagonistic loss function. To further verify the advantage of the proposed antagonistic loss, we compare the performance of three configurations: not using the sparsity loss, using the sparsity loss, and using our antagonistic loss (our method) on the ShanghaiTech dataset, the results are presented in Table 7. We can see that the antagonistic loss can obviously improve model performance, whereas the sparsity loss results in a decrease in model performance.

Table 6. Ablation study on parameter k in MTA on ShanghaiTech.

Model	Hyperparameter k	AUC (%)
Baseline	**–**	**93.8**
Baseline+MTA	3	94.4
	5	**94.8**
	7	94.5
	9	93.9
	11	93.3
	13	93.3
	15	93.0

To delve deeper into why the sparsity loss causes model performance degradation, we train models employing the antagonistic loss and the sparsity loss respectively on the ShanghaiTech dataset, and illustrate their loss curves in Fig. 5, from which we can see a rapid drop of the antagonistic loss as the training goes, conforming to our expectation, whereas the sparsity loss curve exhibits an upward trend, which suggests a gradual increase in the average anomaly score of abnormal videos during the training. This indicates the unsuitability of the

sparsity loss for model training, and further indicates that the proportion of abnormal instances in abnormal videos should not be ignored.

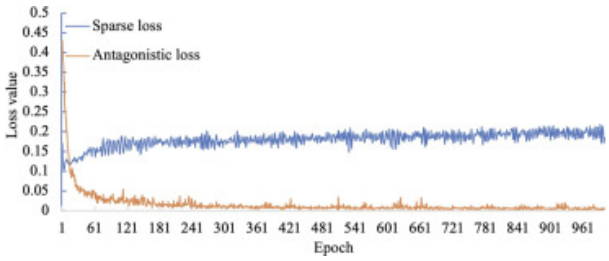

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Fig. 5. The loss curves in model training when using (a) the sparsity loss and (b) our antagonistic loss.

Table 7. Ablation study on loss functions in our method on ShanghaiTech.

Model	AUC (%)-SH
Ours ($loss_{AIS} + loss_{smooth}$)	95.4
Ours ($loss_{AIS} + loss_{smooth} + loss_{sparse}$)	94.8
Ours ($loss_{AIS} + loss_{smooth} + loss_{antagonistic}$)	95.9

4.6. Visual analysis

Visualization of test results. Here, we visualize the test results of our method and the baseline model on the ShanghaiTech and UCF-Crime datasets in Fig. 6, Fig. 7 respectively, to further demonstrate the performance of our method.

In Fig. 6(a), the anomalous event is a man falling to the ground, and in Fig. 6(b) the anomalous event is someone playing skateboard on pavement. However, in Fig. 6(c) and (d), there is no anomaly.

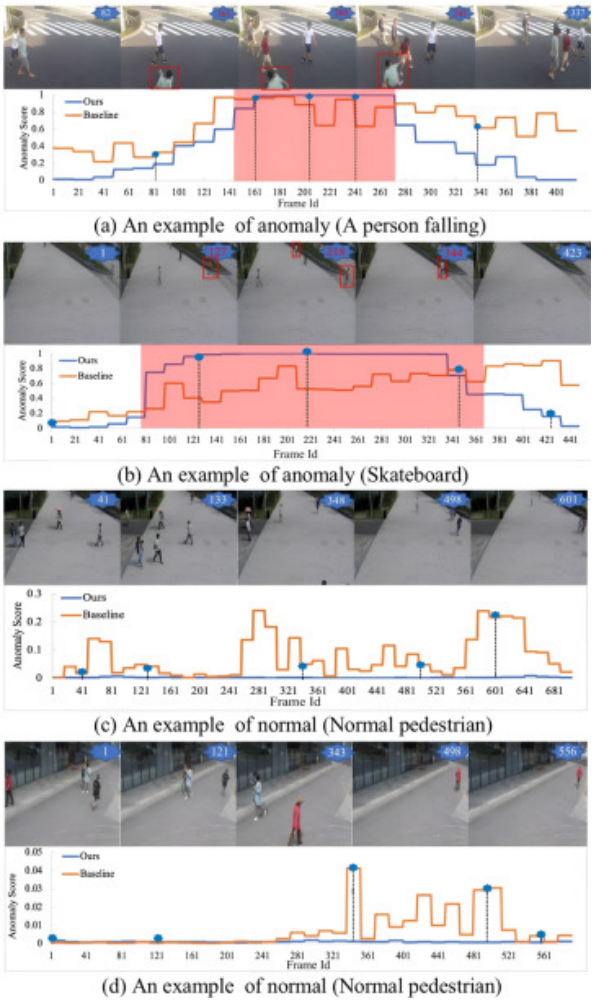

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Fig. 6. Visualization of test results of our method Light-WVAD and the baseline model on the ShanghaiTech dataset. The pink area denotes the time period that anomalies occur in the video, and the numerical labels on the line correspond to the labels in the video frames. In the anomalous video screenshots, the numbers are highlighted in red, and the anomalous objects are marked with red boxes in the frames.

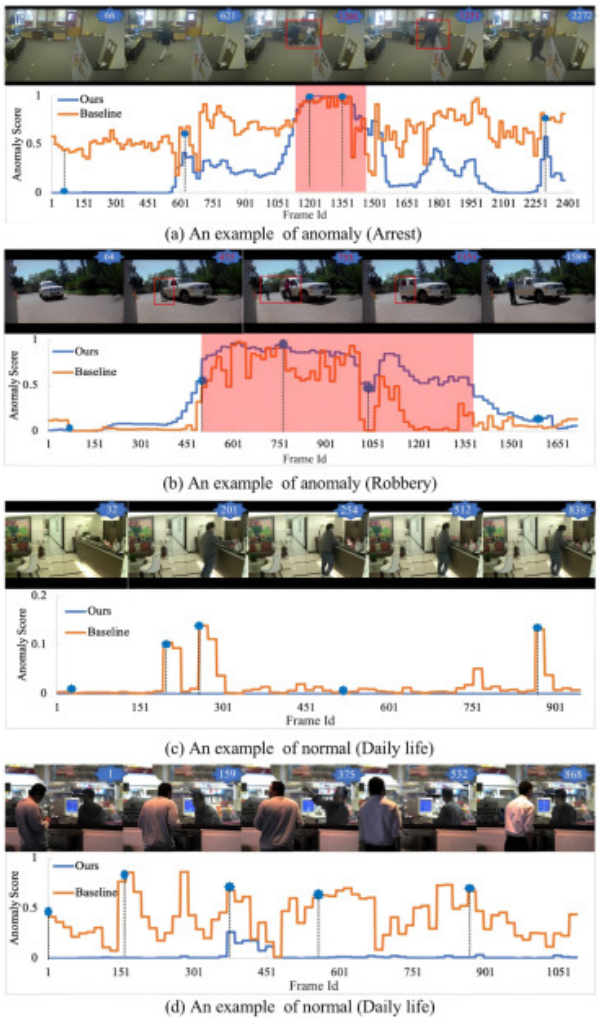

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Fig. 7. Visualization of test results of our method Light-WVAD and the baseline model on the UCF-Crime dataset. The pink area denotes the time period that anomalies occur in the video, and the numerical labels on the line correspond to the labels in the video frames. In the anomalous video screenshots, the numbers are highlighted in red, and the anomalous objects are marked with red boxes in the frames.

As illustrated in Fig. 6(a) and (b), our method can accurately identify the anomalous frames (e.g. No. 161, 205 and 241 in Fig. 6(a), and No. 127, 218 and 344 in Fig. 6(b)), i.e., assigning very high anomaly scores to these anomalous frames, while the baseline model identifys some normal frames as anomaly (e.g. Frame No. 337 in Fig. 6(a) and Frame No. 423 in Fig. 6(b)), and assigns low scores to some anomalies (e.g. Frame 127 in Fig. 6(b)). This indicates that our method has stronger capability of anomaly detection, and is more accurate in detecting the starting and ending of anomalies than the baseline model. Furthermore, Fig. 6(c) and (d) show that our method is more stable on normal video detection than the baseline model by consistently maintaining lower anomaly scores for normal frames.

In Fig. 7(a) and (b), the anomaly is an arrest action and a robbery event, respectively, while in Fig. 7(c) and (d), there is no anomaly.

The visualization results in Fig. 7(a) indicate that our method can accurately identify the anomalous event of police performing an arrest action. Nonetheless, we also observe incorrect responses at Frame No. 621 and No. 2272 in the normal screenshots, which are attributable to the drastic movements of individuals in the scene. Thus, we guess that our model faces challenge in delineating certain normal-anomaly boundaries, primarily due to the absence of detailed annotations. In Fig. 7(b), the anomaly corresponds to a gunman robbing a car,

and a noticeable score fluctuation occurs at Frame No. 1351 in the video, due to the gunman being obscured by the car door, preventing the model from recognizing the anomaly. Nevertheless, outside the scope of the car door's obstruction, our model can still successfully detect the anomaly. Ultimately, as per the visualization results in Fig. 7(c) and (d), our model still exhibits obvious stability on normal videos.

Detection performance of each anomalous class in the UCF-Crime dataset. Fig. 8 presents the AUC of our method on each anomalous class in the UCF-Crime dataset. Compared to RTFM and the baseline, our method yields superior or equivalent detection accuracy across 11 anomaly classes. Specifically, AUC is improved over 12% for the "Assault" and "Stealing" classes. Typically, without long-term analysis of object and human motion, detecting these two types of anomalies is very difficult. Nevertheless, our method achieves high detection accuracy, indicating that our MTA module is effective. Similar to the other methods, our model exhibits suboptimal performance on the "Explosion", "Road Accidents", "Vandalism", and "Abuse" classes. For the sudden anomalies without enough warning signals, they remains a challenge for video anomaly detection.

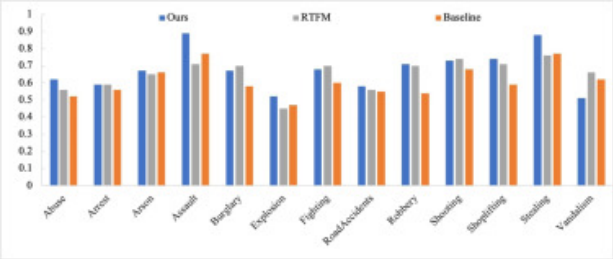

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Fig. 8. AUC results of three methods on different anomalous classes in the UCF-Crime dataset.

4.7. Strengths and weaknesses

The lightweight weakly-supervised video anomaly detection method proposed in this paper successfully reduces the model parameters to only 0.56% of the existing method RTFM through an adaptive instance selection strategy and a multi-level temporal attention module. Additionally, the proposed optimization module achieves significant performance improvement across several non-lightweight methods. This innovation markedly enhances the model's deployability, making it particularly suitable for resource-constrained edge computing scenarios, while performance is maintained with a high AUC score. However, the performance of this model has not yet reached that of non-lightweight methods. Therefore, future work will explore distillation learning to further develop more competitive lightweight approaches.

5. Conclusion

This study develops a lightweight weakly-supervised video anomaly detection method that can effectively addresses the uncertainty and high-parameter issues associated with the existing WVAD methods. Compared with existing lightweight models, Light-WVAD has the smallest number of parameters and the best performance. However, predicting sudden anomalies without warning signals is still a serious challenge for current video anomaly detection methods. And the predicted results on different anomalous classes in the UCF-Crime dataset also show that our model is not effective enough in detecting sudden anomaly classes. In the future, we plan to employ multi-modal techniques to enhance the model's understanding of video content, and thus develop more effective video anomaly detection models.

CRediT authorship contribution statement

Yang Wang: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiaogen Zhou:** Writing – review & editing, Formal analysis, Data curation. **Jihong Guan:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service, or company that could be construed as influencing the position presented in, or the review of, the manuscript.

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Recommended articles

Data availability

No data was used for the research described in the article.

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